INVESTMENT & POLICY CONCEPT · INDIA · 2026

The Grain Value Loop

An integrated rice biorefinery model that turns the byproducts India's government already procures — and currently gives away — into edible oil, food-grade derivatives, energy, and exports. Feedstock the state owns; demand the state can contract.

IMPORT SUBSTITUTION	BRAN GIVEN AWAY TODAY	CAPITAL SUBSIDY
₹21,000 cr	**₹7,650 cr**	**35–50%**
per year, across crops	in free CMR byproduct	PMFME · PMKSY · PLISFPI

FCI–warehouse–MSPvsMSV · figures verified against DFPD, DGCI&S, MoFPI, real project DPRs

THE PARADOX

India imports what it grows — and gifts the feedstock

The government procures ~550 LMT of rice and ~300 LMT of wheat a year, then hands the milling byproducts to private millers for free and imports the finished derivative.

WHAT THE STATE DOES TODAY

- Procures paddy at MSP, hands it to CMR millers

WHAT THE STATE IMPORTS

- **56% of edible oil** (~₹1.65 lakh cr/yr) — while #1 in rice-bran oil

- Millers keep **~180 LMT husk + ~65 LMT bran** — the ₹7,650 cr byproduct giveaway
- Sells surplus grain raw (OMSS) at ~2–4% trader margin, often at a loss

- **67 LMT of pulses**, vital wheat gluten, specialty starch, Vitamin-E, silica
- The *derivative* — while holding the raw material

> World's #1 rice-bran-oil producer, yet imports more than half its cooking oil. The same pattern repeats for gluten, starch, protein, and nutraceuticals across every crop the state procures.

Sources: DFPD Annual Report 2024–25 · DGCI&S FY26 · NITI Aayog edible-oil

THE INSIGHT

Value density rises ~1,000× from grain to derivative

The margin is in the cascade, not the commodity — the sugarcane model (sugar + ethanol + power + CBG) ported to grain.

Paddy
₹24/kg (MSP)

→

Milled rice + bran + husk
the split the state gives away

→

Rice-bran oil
₹105/kg

→

Oryzanol / Vit-E
₹1,200–17,700/kg

COMMODITY FLOOR

₹7–40/kg

flour, feed, husk, power — where the state operates now

INDUSTRIAL MID

₹40–110/kg

RBO, silica, starch, gluten, dal, besan

MOLECULAR TAIL

₹350–17,700/kg

wax, protein, oryzanol, tocotrienols

The state captures the bottom tier and gifts the top three. Source: byproduct product-tree (39 products)

THE PROJECT

An integrated rice biorefinery — one complex, many streams

A cooperative/PPP-hosted complex, sited in an operational Mega Food Park, that takes government CMR paddy and runs the full cascade instead of milling and selling raw:

Rice mill · Bran stabilisation · RBO solvent extraction · RBO refinery · Oryzanol / wax recovery · Husk-fired power · Husk-ash → silica · Broken rice → poha/murmura · Fortified rice kernels · DORB → feed

WHY INTEGRATED

- **Husk powers the process** (already standard mill practice) — energy self-sufficient
- **DORB feed credit** makes the oil cheap (₹60 vs ₹105/kg)
- One feedstock stream → 8+ revenue lines
- The distillery/mill is the natural aggregation host

> Edible oil first (import substitution at ₹105/kg), fuel only from the non-edible residual — India must diverge from the US "distillers-oil-to-biodiesel" model.

Grounded in NCDC cooperative rice-mill DPR + SEA (30 MT existing crush capacity)

RAW MATERIAL · PRICES & SOURCES

Feedstock the state already owns

INPUT	INDICATIVE PRICE	SOURCE / BASIS

Paddy (common)	₹2,369/qtl	MSP KMS 2025-26 · govt CMR procurement (~832 LMT)
Rice bran	₹2,000/qtl (₹20/kg)	NCDC DPR · to oil extractors; 16–22% oil
Rice husk	₹2,000/MT	NCDC DPR · fuel-grade; ~20% of paddy
Broken rice	₹1,500/qtl	NCDC DPR · poha/starch/ethanol feedstock
De-oiled bran (DORB)	₹15/kg (credit)	feed byproduct — the credit that makes RBO viable

AGGREGATION MODEL

PACS / FPO / cooperative societies — *already the paddy-procurement centres* — aggregate paddy + bran. The feedstock chain exists; it is currently dispersed to 100,000+ private mills.

SCALE AVAILABLE TO GOVERNMENT

~832 LMT CMR paddy → ~40 LMT government-touchable bran (41% of national) → ~6.4 LMT RBO at full capture.

Prices: NCDC "Establishment of Rice Mill" DPR (Chhattisgarh) · DFPD MSP · vintage noted

UNIT ECONOMICS · FROM REAL DPRS

Margin and IRR — ranked, government-DPR-backed

PRODUCT	NET MARGIN	IRR	DPR SOURCE
Poha / murmura	10–15%	<3y payback	NIFTEM-T PMFME ✓
RBO solvent extraction	8–15%	–	IMARC/EIRI
Rice mill (anchor)	8.5–15%	19%	NCDC ✓
Maize wet-milling (starch)	12–20%	22–28%	industry
Dal mill	~5.3%	25–31%	RACP + NIFTEM ✓
Roller flour mill	3–5.5%	20–29%	RACP ✓

Margin ≠ return. The staples show thin 3–5% margins but **19–31% IRR** — low capex, fast turnover. The "process, don't sell raw" case rests on return-on-capital, not margin-on-sales.

Sources: NCDC · Rajasthan RACP (Grant Thornton) · NIFTEM-Thanjavur PMFME · IMARC — verified/reported labelled

SCHEMES · THE FUNDING STACK

35–50% of capital comes from MoFPI

PMFME

35%

credit-linked capital subsidy, max ₹10 L/unit · **micro** units (RBO, dal, poha, besan)

PMKSY

35–50%

grants for clusters, cold chain, agro-processing (50% NE/difficult) · **the integrated complex**

PLISFPI

6-yr

incentive on incremental sales + branding-abroad · **large branded** (RBO brand, exports)

MEGA FOOD PARK INFRASTRUCTURE

23 operational parks (of 41), ~₹120 cr each with ~₹50 cr PMKSY grant — ready common facilities (power, cold chain, warehousing, effluent) to plug into, not build.

THE SUBSIDY IS A MARGIN MULTIPLIER

Anchor DPR: ₹80 L cost → ₹16 L subsidy + ₹52 L loan + ₹12 L own. The 19% project IRR becomes a much higher return on the ₹12 L actually invested. Capital-heavy units (RBO, silica, wet-milling) gain the most from a 35–50% grant.

Sources: pmfme.mofpi.gov.in · PMKSY FAQs · mofpi.gov.in/PLISFPI · MoFPI 41-MFP list (31.01.2023)

MARKETS · THREE DEMAND TIERS

Demand the government can contract

CAPTIVE · SUREST

- **Fortified rice kernels — ₹17,082 cr program** to 2028, 100% GoI-funded; ~3.5–4 LMT/yr standing demand
- Fortified atta · Vitamin-E fortificant from own bran

DOMESTIC GROWTH

- **RBO** — India #1 market, $1.39 bn; +exports $104.5 mn
- Poha/murmura (snacks 14.5% CAGR) · besan
- Maize starch $4.1 bn · wheat gluten

EXPORT-LED

- **Basmati ₹50,000 cr** (FY25, +15.7%, ₹82/kg)
- Plant protein (14% CAGR) · spices ₹39,994 cr · green silica (Tata ₹775 cr)

> The offtake is **contractable, not consumer-dependent**: food-processing supply agreements + fortification mandates + institutional (PDS/ICDS) offtake form a demand backbone independent of the retail shelf — the ethanol-blending-mandate lever, applied to processed grain.

Sources: APEDA FY25 (basmati ₹50k cr) · DFPD FRK program · DGCI&S · industry market reports — verified/reported labelled

THE PRIZE · IMPORT SUBSTITUTION

₹21,000 cr/yr across the crops the state holds

STREAM	IMPORT ₹/T (DGCI&S)	SAVINGS ₹ CR/YR	OFFTAKE
Rice bran oil	97,970	7,544	frying/HoReCa + fortification
Pulses (dal margin)	50,888–74,163	~7,000	dal millers + PDS
Cottonseed oil	98,766	1,975	food processing
Maize starch	37,170	1,115	food/pharma/paper

Husk → silica	86,730	867	tyre/industrial
Protein · gluten · Vit-E · lecithin	115k–17.7m	~1,900	processors + export

HIGH-CONFIDENCE TIERS

₹18–21k cr

oil + pulses + industrial mid

ON FEEDSTOCK THE STATE

already owns

via CMR procurement

THROUGH OFFTAKE IT CAN

contract

not hope for

Source: consolidated-savings-by-crop · DGCI&S FY26 import unit values · illustrative, additive

A CONCRETE MSV CASE · WHEAT → PASTA

Durum wheat: the fastest-growing value-add India under-grows

The same "import the derivative, hold the feedstock" pattern — in wheat. India is **wheat-surplus** and sells it cheap via OMSS (~₹2,550/qtl), yet:

- Pasta market **$1.21 bn, growing ~14–16%/yr** — the fastest wheat-product segment
- India is the **world's #2 durum-semolina importer** — pasta needs durum (hard) wheat, which India barely grows (MP's Malwa belt, ~1.5–2 MT)
- DGCI&S: pasta imports **+24%**, biscuit imports **+28%**, gluten **+35%** — the value-added tier's imports are building

THE DOUBLE PLAY (DURUM MSV)

- **Procure durum by variety** — a variety-specific MSV that de-risks durum cultivation (the wheat parallel to the oilseed-MSP production lever)
- **Mill semolina → pasta** on OMSS/durum wheat, in a Mega Food Park
- Substitutes **both** the pasta imports AND the durum-wheat imports

- **Export prize:** India ships only **$63 mn** of pasta into a global import trade of **US $1.62 bn · Germany $1.24 bn** — under 4% of the US market, on cheap OMSS wheat Italy can't match

- + vital gluten (13,000 t/yr imported) from the same wheat wet-mill

> India is a net *exporter* of biscuits (Parle-G, 180 countries) — so the target is the **premium/pasta/durum tier** where imports are growing, not the mass market. Import substitution is already visible: durum-pasta imports fell −25% YoY.

Sources: DGCI&S TradeStat EIDB (exact 8-digit HS, FY25–26 prov.) · industry market sizes · report/wheat-value-added-products.md

A CONCRETE CASE · EDIBLE OIL FROM FCI/CMR STOCK REUTILISATION

The oil India imports for ₹1.68 lakh cr — from grain the state already holds

India imports **168 LMT of edible oil — US$19.3 bn, ₹1.68 lakh cr** a year (57% of demand), landing at **~₹100/kg crude** (palm ₹96). Yet the 24 MSP crops **are** the domestic oil-feedstock base — and FCI's own procured paddy throws off the cheapest oil in the country as a byproduct:

- **Rice bran oil** — bran from the ~130 MT of paddy under CMR; **₹60/kg ex-mill vs ₹105 import parity = ₹45/kg edge**. India is the world's #1 RBO producer — and hands the bran to millers **free** (the ₹7,650 cr giveaway)
- RBO is also the *better* oil: γ-oryzanol, tocotrienols, phytosterols; cholesterol-

IMPORT-SUBSTITUTION SAVINGS ESTIMATE

STREAM	₹/KG EDGE	₹ CR/YR
Rice bran oil (7.7 LMT)	45	7,544
Cottonseed oil (2 LMT)	~10	1,975
Soy/sun crush — cake + forex*	system	~1,500
Capturable now		**~₹11,000 cr/yr**

*at-parity oils save via the oil-cake co-product imports don't carry + forex retained, not a retail-price cut.

lowering; high-heat stable; non-GMO (ICRBO-2016)

- **Cottonseed oil** — byproduct of cotton (an MSP crop): crush margin + cake

> Honest tiering: **~₹11,000 cr/yr is capturable now** from byproduct oils on feedstock the state already procures. The **production lever** (NMEO-OS oilseeds 39→69.7 MT) is the ceiling — closing even half the 168-LMT gap replaces **~₹84,000 cr** of imports over 5–7 yrs — but mustard/groundnut are premium oils, a *volume/self-sufficiency* lever, never an in-year cheaper-cooking-oil promise.

Sources: DGCI&S TradeStat EIDB (168 LMT/₹1.68 L cr) · DFPD edible-oil scenario · SEA (16.3 MT/₹1.61 L cr) · consolidated-savings-by-crop · report/msp-crops-refined-oil-source.md

THE PDS · ONORC CASE · THE DEMAND RAIL ALREADY EXISTS

₹2 lakh cr already spent, 80 cr served — redirect the oil spend to domestic jobs

DOMESTIC JOBS

FAIR PRICE SHOPS

5.43 lakh

99.8% ePoS-automated · dealer + staff livelihoods, every district

RICE MILLS

75k–100k

10–50 workers each · TN 3,399 · AP 1,913 · WB 1,873

SOLVENT-EXTRACTION PLANTS

PRODUCTION VOLUMES

Rice procured	~520 LMT
Wheat procured	300 LMT
Paddy milled/yr	~850 LMT
Rice bran oil	~10 LMT
Domestic edible oil	122 LMT
Edible oil imported	168 LMT

STATE EXPENDITURE

CENTRAL FOOD SUBSIDY FY25

₹2.05 L cr

free rice/wheat to ~80 cr (NFSA+PMGKAY) + ₹7,075 cr FPS margins

TN SPECIAL-PDS OIL+DAL

₹3,800 cr

palmolein at ₹25/L on ~₹65/L subsidy — **on an**

~350

875 SEA members · 30 MT processing capacity

Oilseeds grown	40.99 MT

The milling that makes free PDS rice already throws off the bran for domestic oil.

import (AP/KA/TG/WB run their own)

ONORC COVERAGE

36/36

States/UTs · 20.54 cr cards · portable, deduplicated

> The state already spends **₹2 lakh cr+** and runs **5.43 lakh shops** to feed 80 cr people — the demand rail is built. **ONORC** turns it into one portable, deduplicated national offtake: redirect the state *oil* subsidy from imported palmolein to **domestic RBO**, and the same rupee funds rice-mill, solvent-plant and oilseed-farm jobs instead of foreign crushers — import substitution and employment on infrastructure that already exists.

Sources: PRS Demand-for-Grants 2024-25 (Food & PD) · DFPD/PIB Year-End 2024 · TNCSC · DCMSME · SEA of India · data/pds_onorc_jobs_volumes_spend.csv

THE MODEL · WHY GOVERNMENT-COOPERATIVE-PPP

Ring-fence the commercial arm; contract the offtake

STRUCTURE

- **PACS/FPO** aggregates paddy + bran (already the procurement centre)
- **Private partner** builds + runs RBO extraction, silica, power (the capital + know-how)
- **State** provides VGF/PMKSY grant + a fortified-rice / RBO **offtake guarantee**

WHY NOT LEAVE IT TO PRIVATE MILLERS

- 100,000 mills do basic CMR — but won't invest in bran stabilisation, RBO, silica, cogen
- The integrated **value-added** mill is the PPP white space (verified: largely untried)
- The offtake guarantee is the revenue backbone — mirrors PEG's silo-hire guarantee

- Ring-fenced from MSP/buffer policy — the COFCO lesson, not the STC/Thailand failure

> History's warning (verified): STC→PEC→NCEL failed by trading naked; CWC, AMUL, Kendriya Bhandar succeeded by infrastructure / farmer-ownership / ring-fencing. This is an *infrastructure + processing* play, not a trading arm.

Sources: PPP rice-mills · PSU-agri-ventures-history · agri-trader-playbook

ILLUSTRATIVE PILOT

One integrated complex, in an operational food park

ANCHOR: RICE MILL **₹80 L** 1 TPH, 19% IRR (NCDC DPR)	+ RBO EXTRACTION **₹15–40 cr** 8–15% margin, the ₹7,544 cr anchor	+ POHA/MURMURA LINE **₹21 L** 10–15% margin, <3y payback
+ HUSK POWER/SILICA **cluster** energy self-sufficiency + tyre-grade silica	CAPITAL SUBSIDY **35–50%** PMKSY cluster + PMFME units	OFFTAKE SECURED BY **contract** FRK/PDS + food processors + fortification

> The proof-points already exist as verified DPRs and operational parks. The novelty is **integration + the offtake guarantee** — assembling proven units into one government-anchored complex that captures the cascade instead of gifting it.

All component economics from real project DPRs; integration is the un-built step

WHY IT MATTERS · DOUBLING FARMERS' INCOME

Real farm income is ~a third of the way to doubling

The 2016 goal: double farmers' *real* income by 2022-23. Deflated by CPI-AL (the farm household's own cost basket, MoSPI), the **Real Farm Income Index** reads:

2016-17
RFII 99

→

2021-22
RFII 131

→

Target
200 = doubling

Achieved ~5.7%/yr real vs the 10.4%/yr needed — reaching 200 slips to ~2030.

THE LEVER THE GAP NEEDS

- Crop income is only **37%** of farm income — price alone can't double it
- The other 63%: wages, livestock, **non-farm/value-addition**
- Cutting *cost* defends real income against inflation the nominal headline hides
- Exactly the byproduct-cascade + processing-margin this project builds

> RFII is a monitorable metric: CPI-AL deflator refreshes monthly (MoSPI), income on each SAS/NAFIS survey.

Sources: NABARD NAFIS 2021–22 · NSSO SAS · MoSPI CPIALRL · Dalwai Committee (DFI)

THE FARM-SIDE LOOP · FUEL CHOICE

Diesel→CNG + residue: the inflation-proof income lever

FUEL SHIFT (DIESEL→CNG)

+₹5,400

FULL CBG LOOP (+RESIDUE)

+₹9,750

MILL FUEL CHOICE

₹1 vs ₹10.45

/yr per 1.5-ha paddy household · 24% on tractor fuel · +3.3 RFII pts

/yr · residue→CBG + FOM slurry · +5.9 RFII pts · closes 9% of the doubling gap

husk/CBG per kWh vs grid · ₹2.7-7.9 cr/yr on the mill's EBITDA

> **The loop closes:** crop → residue → CBG → fuels the tractor (24% cheaper, inflation-proof) AND the mill's boiler → slurry to the field. Diesel is a volatile imported input; CBG from the farmer's own residue defends the *real* income the doubling metric measures.

CACP A2+FL · PIB CNG-tractor (PRID 1697555) · SATAT CBG · TN diesel/CNG Jul 2026

THE ONE-LINE CASE

Stop gifting the byproduct. Contract the demand. Capture the cascade.

India is the world's #1 producer of the raw byproduct and a net importer of the refined derivative — for oil, gluten, starch, protein, and Vitamin-E alike. The feedstock is procured, the subsidy is 35–50%, the parks are built, the DPRs are proven. What's missing is the integrated complex and the offtake guarantee. That is the project.

SAVINGS	BEST DPR MARGIN	BEST DPR IRR
₹21k cr/yr	**poha 10–15%**	**dal 25–31%**

Full analysis, data & sources: github.com/herrrickshaw/FCI-warehouse-MSPvsMSV · research synthesis, not investment advice